UNITED STATES PROVISIONAL PATENT APPLICATION

QUASICRYSTALLINE STRUCTURAL ELEMENT FOR PARAMETRIC FREQUENCY CASCADE WITH INTEGRATED THERMOELECTRIC ENERGY HARVEST AND DIRECTED ACOUSTIC EMISSION

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Related Applications: This application is related to U.S. Provisional Application No. 63/995,401 filed March 3, 2026, titled "Quasicrystalline Thermoelectric Coating System with Fibonacci-Sequence Layer Architecture for Adaptive Sensing and Energy Harvesting," and to a concurrently filed provisional application titled "Adaptive Cutting System with Quasicrystalline Thermoelectric Sensing Coating for Real-Time Material Identification and Process Optimization," the disclosures of which are incorporated herein by reference in their entirety.

ABSTRACT

A structural element comprising a phi-graded quasicrystalline material arranged in multiple layers with golden-ratio impedance grading, wherein the element simultaneously bears mechanical load, converts thermal energy to electricity via integrated thermoelectric electrode layers, and generates directed acoustic or electromagnetic emission through parametric frequency cascade. The nonlinear properties of the quasicrystalline medium, combined with the golden-ratio spacing of layer boundaries, create a resonant frequency comb wherein difference-frequency mixing products at each boundary land precisely on lower-order resonant modes of the structure. This enables coherent energy transfer from high-frequency broadband input (mechanical vibration, thermal noise) to arbitrarily low-frequency directed emission via a cascade of self-reinforcing parametric down-conversion steps. When integrated into a propulsion system comprising a plurality of thrust-generating engines, the engine mounts and structural thrust elements serve simultaneously as the nonlinear cascade medium, thermoelectric generator, and phased-array acoustic antenna. Applications include launch vehicle weight reduction through directed acoustic emission, vibration energy harvesting, thermal power generation, and geological drilling systems.

FIELD OF THE INVENTION

The present invention relates to structural elements composed of quasicrystalline materials arranged in phi-graded multilayer architectures that simultaneously serve as load-bearing structures, thermoelectric generators, nonlinear acoustic/phononic media, and directed emission antennas. More particularly, the invention relates to systems and methods for converting broadband mechanical vibration and thermal energy into coherent directed emission at controllable frequencies through parametric frequency cascade in golden-ratio-structured nonlinear media.

BACKGROUND OF THE INVENTION

Modern propulsion systems, industrial machinery, drilling equipment, and power generation systems produce enormous quantities of waste energy in the form of mechanical vibration and heat. In launch vehicles, for example, chemical rocket engines convert approximately 0.5-3% of their total mechanical power output into acoustic energy, representing gigawatts of power that is currently treated exclusively as a structural hazard to be mitigated through damping, isolation, and vibroacoustic hardening (NASA-STD-5002). Additional thermal energy from exhaust radiation, aerodynamic heating, and conduction through engine mounts further loads vehicle structures without contributing to propulsion.

Acoustic levitation is a known phenomenon wherein acoustic radiation pressure supports objects against gravity. However, conventional acoustic levitation is limited to objects much smaller than the acoustic wavelength and forces on the order of millinewtons. The fundamental limitation is the impedance mismatch between structural materials (acoustic impedance Z approximately 46 million Rayl for steel) and atmosphere (Z approximately 420 Rayl), resulting in less than 0.004% transmission of acoustic energy from bare structural surfaces.

Nonlinear wave mixing in structured media is well-established in optics, where optical frequency combs (Nobel Prize in Physics, 2005) achieve coherent frequency conversion spanning many decades. Parametric amplification in nonlinear crystals enables efficient conversion between frequencies with gains exceeding 10^6. However, the application of these principles to acoustic/phononic systems in structural elements, particularly using quasicrystalline media with golden-ratio architecture, has not been explored.

Quasicrystalline materials possess unique phonon spectra arising from their aperiodic lattice structure. Unlike periodic crystals with integer-harmonic phonon modes, quasicrystals exhibit phonon modes spaced by the golden ratio phi = (1 + sqrt(5))/2 approximately 1.618. This phi-spacing has the property that difference-frequency mixing products between adjacent modes land precisely on lower-order modes of the same structure, enabling lossless cascading of energy from high to low frequencies. The present invention exploits this property to create structural elements that convert broadband vibrational and thermal energy into coherent directed emission at controllable frequencies.

Prior art in thermoelectric energy harvesting (U.S. Provisional Application No. 63/995,401) established quasicrystalline coatings with Fibonacci-sequence layer architecture for adaptive sensing and energy harvesting. The present invention extends that architecture from thin-film coatings to bulk structural elements, from passive sensing to active parametric frequency conversion, and from individual components to integrated propulsion and energy systems.

SUMMARY OF THE INVENTION

The present invention provides a structural element comprising phi-graded quasicrystalline material arranged in N layers (preferably 13) with golden-ratio impedance grading between adjacent layers. The structural element incorporates integrated thermoelectric electrode layers (preferably at positions corresponding to the 2nd, 8th, and 12th layers) that serve bidirectionally as thermal energy harvesters (Seebeck mode) and active parametric pump sources (Peltier/piezoelectric mode).

When the structural element is subjected to broadband mechanical vibration and thermal gradients (as occurs naturally in propulsion systems, drilling equipment, and industrial machinery), the phi-graded quasicrystalline medium performs simultaneous functions: (a) bearing the mechanical loads of the application; (b) converting

thermal energy to electrical energy through the integrated thermoelectric layers; (c) sorting broadband vibrational energy onto a phi-spaced resonant frequency comb; (d) cascading energy from high frequencies to low frequencies through nonlinear difference-frequency mixing at each layer boundary, with each mixing product landing on a structural resonance; (e) amplifying the cascade through parametric gain provided by the thermoelectrically-powered electrode layers; and (f) radiating coherent directed acoustic emission from the element surfaces, where the compress-hinge-expand gradient of the phi-architecture determines the directionality of emission.

In a preferred embodiment for propulsion systems, a plurality of thrust-generating engines are mounted to a structural thrust element composed of the phi-graded quasicrystalline material. The engine mounts simultaneously inject broadband acoustic energy into the nonlinear medium and provide thermal gradients for thermoelectric generation. The engines are arranged in a phased-array configuration (preferably a concentric ring pattern such as 20-10-3) enabling coherent beam-forming of the low-frequency directed emission. Cryogenic propellant may be routed through channels within the quasicrystalline element, simultaneously cooling the nonlinear medium (enhancing parametric efficiency), pre-heating the propellant (improving specific impulse), and maximizing the thermal gradient for thermoelectric generation.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

1. Phi-Graded Quasicrystalline Structural Element

Referring to FIG. 1, a structural element (100) comprises 13 zones of quasicrystalline material arranged with phi-graded dimensions. Zone 1 (the largest) has a characteristic dimension d_1, and each subsequent zone has dimension $d_n = d_1 / phi^{n-1}$. The total length of the element is the sum of all zone dimensions: $L = d_1 \times (1 + 1/phi + 1/phi^2 + ... + 1/phi^{12}) = d_1 \times phi^2 \times (1 - 1/phi^{13}) / (1 - 1/phi)$ approximately equal to $d_1 \times 2.618 \times 1.998$ approximately equal to $5.23 \times d_1$.

Each zone comprises quasicrystalline material having icosahedral symmetry, preferably of the Al-Cu-Fe family (approximate composition $Al_{63}Cu_{25}Fe_{12}$) or related ternary/quaternary quasicrystalline phases. The acoustic impedance of each zone is graded from high impedance at the load-bearing (substrate) end to low impedance at the emission (surface) end. This grading is achieved through compositional variation, porosity control, or incorporation of secondary phases within the quasicrystalline matrix.

Zones 2, 8, and 12 (numbering from the highest-impedance end) incorporate electrode layers comprising copper nanowire (CuNW) networks embedded within the quasicrystalline matrix. These electrode layers serve bidirectionally: in Seebeck mode, thermal gradients across the quasicrystalline zones generate voltage for energy harvest; in Peltier/piezoelectric mode, applied voltage drives mechanical displacement for parametric pumping of the cascade.

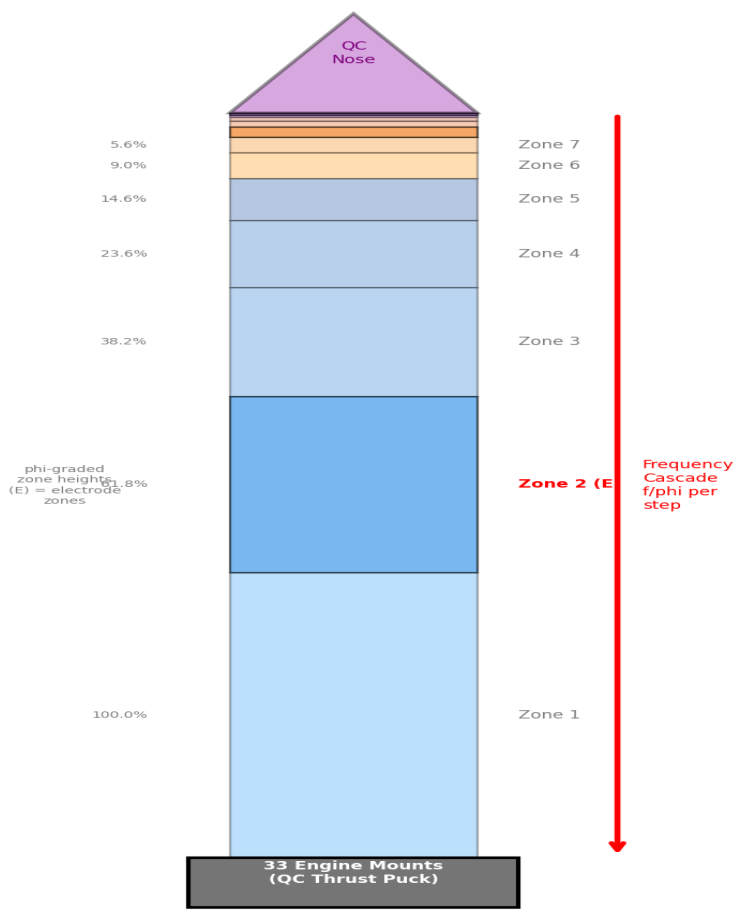

FIG. 1 — Vehicle as 13-Zone Parametric Cascade Structure

FIG. 1 — Vehicle as 13-Zone Parametric Cascade Structure, showing phi-graded zone heights and electrode (E) positions at zones 2, 8, and 12.

2. Parametric Frequency Cascade Mechanism

Referring to FIG. 2, the phi-graded structure creates a resonant frequency comb with modes at frequencies $f_n = f_0 \times phi^n$ for integer n, where f_0 is the fundamental resonant frequency of the largest zone. The fundamental frequency is determined by the zone dimension and the speed of sound in the quasicrystalline material: $f_0 = c_{QC} / (4 \times d_1)$.

At each boundary between adjacent zones, the impedance discontinuity creates a nonlinear scattering site. When acoustic energy at frequency f_n (resonant mode n) interacts with energy at frequency f_{n+1} (resonant mode n+1) at the boundary, the nonlinear elastic response of the quasicrystalline medium generates difference-frequency products at $f_{diff} = f_{n+1} - f_n = f_0 \times phi^n \times (phi - 1) = f_0 \times phi^{n-1}$, which is precisely the resonant mode at level n-1. This self-similar property of the golden ratio ensures that every mixing product is itself a resonant mode of the structure, preventing energy loss to non-resonant frequencies.

The cascade proceeds stepwise: energy enters at high frequency (broadband vibration from engines or machinery), is sorted onto the phi-spaced comb, and progressively mixes down to lower frequencies at each boundary. After N cascade steps, the output frequency is $f_{out} = f_{in} / phi^N$. For typical structural vibration input (10 Hz to 10 kHz) and a 13-zone element enabling approximately 55-70 cascade steps through multiple internal reflections, the output frequency extends to the sub-millihertz range.

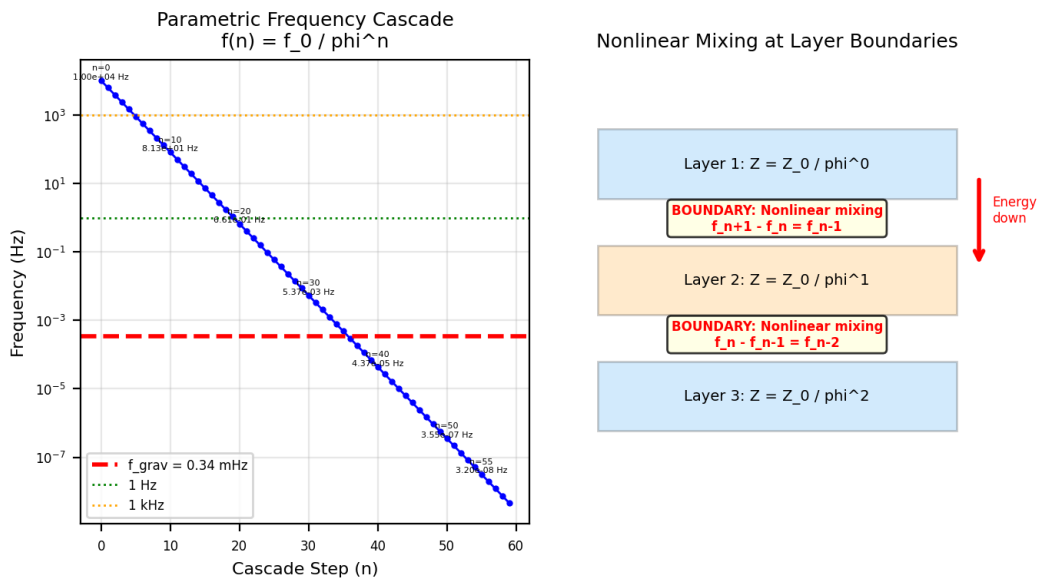

FIG. 2 — (a) Parametric frequency cascade showing frequency vs cascade step number. Each step divides frequency by phi. Earth's gravitational characteristic frequency (0.34 mHz) is reached at approximately step 55. (b) Nonlinear mixing at layer boundaries showing how difference-frequency products land on structural resonances.

3. Parametric Amplification via Thermoelectric Pumping

In a passive nonlinear medium, each cascade step converts only a fraction of the input energy (typically 0.01-1%), and the cascade would attenuate to negligible levels after many steps. The present invention overcomes this limitation through active parametric amplification at the electrode layers.

The thermoelectric electrode layers at positions 2, 8, and 12 operate bidirectionally in an alternating sense-drive cycle. In Seebeck (sense) mode, the electrode measures the local thermal gradient and generates electrical power. At thermal gradients typical of propulsion systems (delta-T of 3,000-3,500 K between exhaust and cryogenic propellant), the Carnot efficiency approaches 97%, and realistic thermoelectric conversion efficiency at ZT = 2-3 yields 30-35% of Carnot, providing substantial electrical power from surfaces as small as 10 m^2.

In Peltier/piezoelectric (drive) mode, the harvested electrical energy is fed back into the quasicrystalline medium as a parametric pump signal at frequencies determined by the controller. The pump signal modulates the local nonlinear coefficient of the medium, enhancing the conversion efficiency at each cascade step. When the pump

power exceeds the parametric gain threshold (determined by the product of pump intensity, nonlinear coefficient, and interaction length), each cascade step provides net amplification rather than attenuation. The signal at each successively lower frequency grows rather than decays, drawing energy from the pump reservoir.

The three electrode positions (2, 8, 12) are distributed through the structure such that the parametric pump reinforces the cascade at low, middle, and high frequency ranges respectively, ensuring continuous amplification across the entire frequency span.

4. Propulsion System Embodiment: QC Thrust Puck

Referring to FIG. 3, in a preferred embodiment for launch vehicle propulsion, the structural thrust element (thrust puck) that bears the mechanical load of all engine thrust forces is composed of or augmented with phi-graded quasicrystalline material. A plurality of thrust-generating engines (preferably 33, arranged in a 20-10-3 concentric ring pattern) are mounted to the thrust element at flanges that serve simultaneously as acoustic injection points and thermal gradient interfaces.

The 13 inner gimbaling engines (10 middle ring + 3 center cluster) correspond to the 13-zone phi-graded structure, with each engine mount interfacing with a different structural zone. The 3 center engines, clocked at pentagonal angles (approximately 108 degrees), correspond to the electrode zones and provide fine phase control for beam-forming of the directed emission. The 10 middle gimbaling engines provide the active cascade zones. The 20 outer fixed engines provide raw thermal and acoustic energy as the pump substrate.

Cryogenic propellant channels are routed through the quasicrystalline thrust element between the exhaust-side and structural-side faces. Liquid methane at approximately 90 K on one face and exhaust products at approximately 3,500 K on the other face create a thermal gradient of approximately 3,410 K across the element, driving the thermoelectric electrode layers at high efficiency. The cryogenic propellant simultaneously cools the nonlinear medium (increasing the nonlinear coefficient and parametric gain) and absorbs waste heat (pre-heating the propellant to improve specific impulse).

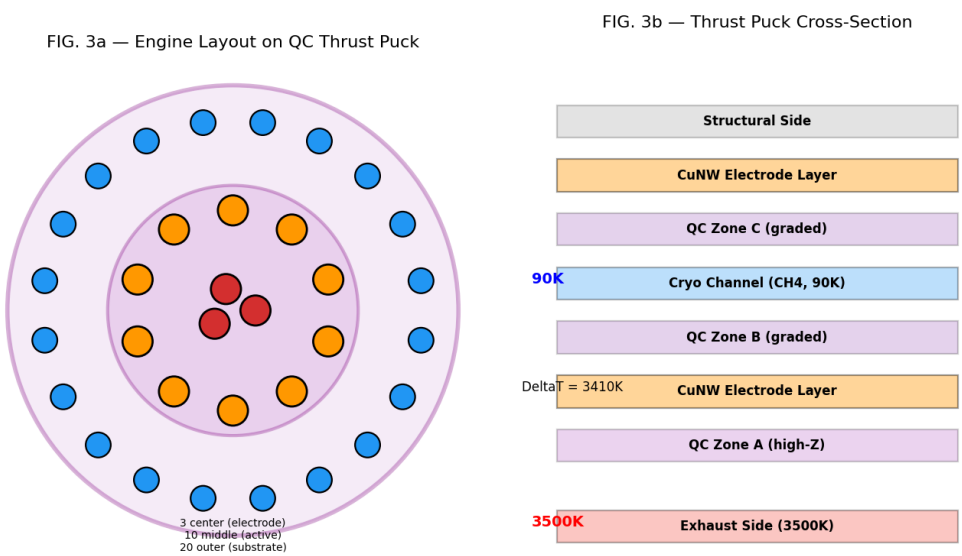

FIG. 3 — (a) Top view of engine layout on QC thrust puck showing 20-10-3 concentric ring pattern with inner QC region highlighted. (b) Cross-section of thrust puck showing QC zones, CuNW electrode layers, and cryogenic propellant channel.

5. Phased Array Beam-Forming

Referring to FIG. 4, the plurality of engine mounts, each equipped with individually controllable piezoelectric or magnetostrictive actuators powered by the thermoelectric harvest, form a phased-array acoustic antenna. By controlling the relative phase of the low-frequency emission from each mount point, the combined radiation pattern is steered and focused. For N coherent array elements, the maximum array gain is N^2 (for amplitude) or N

(for power), concentrating the radiated energy into a narrow beam.

For the 13-element inner array (gimbaling engines), coherent beam-forming concentrates directed emission into a beam with angular width inversely proportional to the array aperture. The 3 center engines provide fine phase correction, the 10 middle engines provide the primary beam shape, and the 20 outer engines provide the broad-spectrum acoustic substrate. A controller adjusts the phase of each element in real time based on feedback from accelerometers and gravimeters mounted on the vehicle structure.

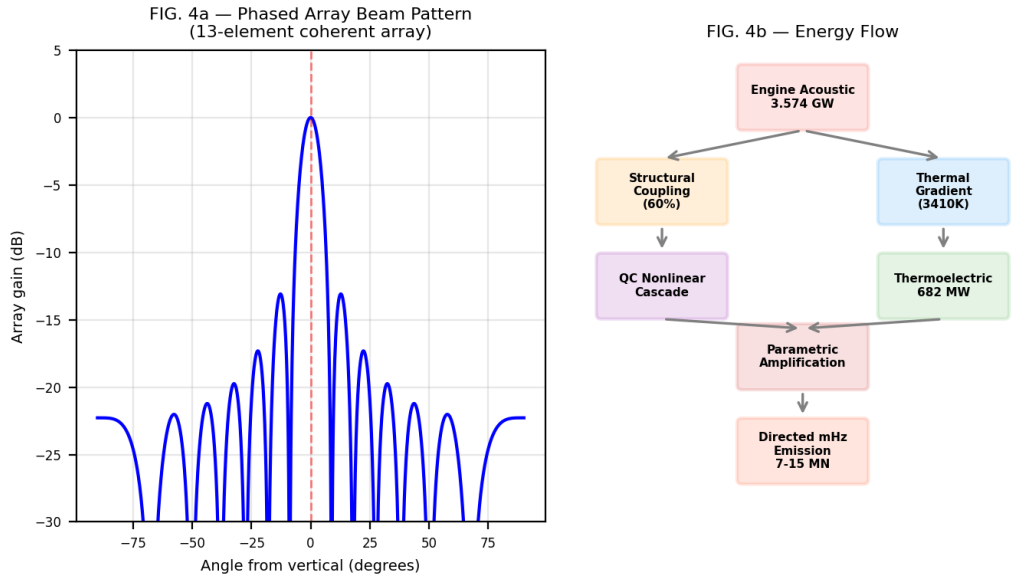

FIG. 4 — (a) Phased array beam pattern for 13-element coherent array showing main lobe and sidelobes. (b) Energy flow diagram from engine acoustic and thermal input through QC nonlinear cascade and thermoelectric harvest to directed emission.

6. Dual-Direction Emission Architecture

In a further preferred embodiment, the quasicrystalline material is applied at both the thrust element (base) and the nose structure (apex) of the vehicle. The compress-hinge-expand gradient of the phi-architecture at the base is oriented with the compress face outward (downward), while the gradient at the nose is oriented with the expand face outward (upward). This dual orientation creates a dipole radiation pattern at the cascade output frequency: preferential emission downward from the base and upward from the nose.

For frequencies at which the vehicle length is much less than the acoustic wavelength (all frequencies below approximately c_{air} / L approximately 3 Hz), the vehicle radiates as a single dipole element rather than as a distributed source. The reaction force from this dipole emission acts on the vehicle center of mass, providing net uplift proportional to the radiated power divided by the speed of sound in the surrounding medium.

At the lowest cascade frequencies (sub-millihertz), the emission wavelength exceeds planetary scales. At these frequencies, the radiation interacts with the ambient gravitational field gradient rather than with the atmospheric medium. The phi-structured emission, if phase-matched to the local gravitational field characteristics, may produce destructive interference with the gravitational gradient, reducing the effective gravitational acceleration experienced by the vehicle.

7. Additional Embodiments

The phi-graded quasicrystalline structural element finds application in any environment combining high mechanical vibration with significant thermal gradients. Geological drilling systems (tunnel boring machines, horizontal directional drilling) experience vibration from cutting action and thermal gradients from frictional heating versus cooling fluid. Industrial turbomachinery (gas turbines, compressors) produces broadband vibration at high power levels with substantial thermal gradients across hot and cold sections. In each case, the phi-graded element converts waste vibrational and thermal energy into useful directed emission or electrical power.

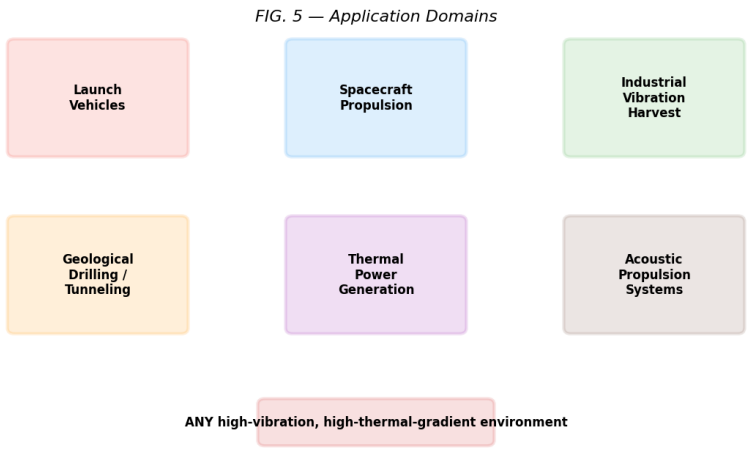

FIG. 5 — Application domains for phi-graded quasicrystalline structural elements.

CLAIMS

What is claimed is:

1. A structural element comprising:

(a) a load-bearing body composed of quasicrystalline material arranged in a plurality of zones, wherein the zones are dimensioned in a geometric sequence with a ratio between adjacent zone dimensions approximately equal to the golden ratio phi = (1 + sqrt(5))/2;

(b) a plurality of thermoelectric electrode layers integrated within the quasicrystalline body at predetermined positions, each electrode layer comprising electrically conductive elements embedded within the quasicrystalline matrix, the electrode layers configured to operate bidirectionally in a Seebeck mode generating voltage from thermal gradients and in a Peltier/piezoelectric mode generating mechanical displacement from applied voltage;

(c) an impedance grading across the plurality of zones from a high-impedance face to a low-impedance face;

wherein the structural element simultaneously bears mechanical load, converts thermal energy to electrical energy via the thermoelectric electrode layers, and generates directed acoustic emission through nonlinear parametric frequency cascade at the zone boundaries, the cascade producing output frequencies lower than input frequencies by a factor of phi per cascade step.

2. The structural element of claim 1, wherein the plurality of zones comprises 13 zones and the thermoelectric electrode layers are positioned at zones 2, 8, and 12, numbered from the highest-impedance face.

3. The structural element of claim 1, wherein the quasicrystalline material comprises an Al-Cu-Fe icosahedral quasicrystalline phase.

4. The structural element of claim 1, wherein the thermoelectric electrode layers comprise copper nanowire networks.

5. The structural element of claim 1, wherein the nonlinear parametric frequency cascade produces a resonant frequency comb with modes at $f_n = f_0 \times phi^n$, and wherein difference-frequency mixing products at each zone boundary land on resonant modes of the structure.

6. The structural element of claim 5, wherein the frequency cascade spans from input frequencies in the range of 1 Hz to 100 kHz to output frequencies in the range of 0.0001 Hz to 1 Hz through a plurality of cascade steps.

7. The structural element of claim 1, further comprising channels for passage of a cooling fluid through the quasicrystalline body, the cooling fluid simultaneously cooling the nonlinear medium and absorbing thermal energy.

8. A propulsion system comprising:

(a) a structural thrust element composed of phi-graded quasicrystalline material as set forth in claim 1;

(b) a plurality of thrust-generating engines mounted to the structural thrust element, the engines injecting broadband acoustic energy and thermal energy into the quasicrystalline material at the engine mounting interfaces;

(c) a controller configured to drive the thermoelectric electrode layers in a parametric amplification mode using electrical energy harvested from the thermal gradient across the structural thrust element;

wherein the structural thrust element converts a portion of the waste acoustic and thermal energy from the engines into directed coherent emission, producing a reaction force on the structural thrust element.

9. The propulsion system of claim 8, wherein the plurality of engines are arranged in a concentric ring pattern comprising an outer ring of fixed engines, a middle ring of gimbaling engines, and a center cluster of gimbaling engines.

10. The propulsion system of claim 9, wherein the center cluster comprises 3 engines and the middle ring comprises 10 engines, the 13 gimbaling engines collectively corresponding to 13 zones of the phi-graded quasicrystalline structural element.

11. The propulsion system of claim 8, wherein the engine mounts are equipped with individually controllable actuators forming a phased-array acoustic antenna, the controller adjusting relative phase between actuators to concentrate directed emission into a beam.

12. The propulsion system of claim 8, further comprising cryogenic propellant channels routed through the structural thrust element, wherein cryogenic propellant at a first temperature flows through the channels while engine exhaust products at a second, higher temperature contact the opposite face of the structural thrust element, creating a thermal gradient driving the thermoelectric electrode layers.

13. The propulsion system of claim 12, wherein the first temperature is approximately 90 K and the second temperature is approximately 3,500 K.

14. The propulsion system of claim 8, wherein the directed coherent emission comprises acoustic emission at frequencies below 1 Hz produced by parametric frequency cascade from broadband engine vibration input.

15. The propulsion system of claim 8, further comprising a second structural element of phi-graded quasicrystalline material at a nose or apex structure of the vehicle, the second structural element having a compress-hinge-expand gradient oriented opposite to the gradient of the structural thrust element, creating a dipole emission pattern.

16. A method of converting waste vibrational and thermal energy into directed emission, comprising:

(a) providing a structural element composed of phi-graded quasicrystalline material with integrated thermoelectric electrode layers;

(b) subjecting the structural element to broadband mechanical vibration from an energy source mechanically coupled to the element;

(c) generating electrical energy from thermal gradients across the structural element using the thermoelectric electrode layers in Seebeck mode;

(d) driving the thermoelectric electrode layers in parametric amplification mode using a portion of the generated electrical energy, thereby enhancing nonlinear mixing efficiency at zone boundaries;

(e) cascading vibrational energy from input frequencies to lower output frequencies through successive difference-frequency mixing at the phi-spaced zone boundaries;

(f) emitting the low-frequency output as directed acoustic emission from the element surface, the directionality determined by the impedance gradient of the phi-graded structure.

17. The method of claim 16, wherein the broadband mechanical vibration is produced by one or more rocket engines, and the directed emission produces a reaction force that partially offsets the gravitational weight of a vehicle to which the structural element is attached.

18. The method of claim 16, wherein the broadband mechanical vibration is produced by a drilling or cutting operation, and the directed emission is used for geological formation characterization or material identification.

19. A structural thrust element for a launch vehicle, comprising a three-dimensional quasicrystalline lattice having Penrose-tiling geometry in at least two dimensions, the lattice arranged as a phi-graded multilayer structure with integrated thermoelectric electrode layers, configured to simultaneously bear the thrust forces of a plurality of rocket engines and convert waste acoustic and thermal energy from the engines into directed coherent acoustic emission through parametric frequency cascade.

20. The structural thrust element of claim 19, wherein the three-dimensional quasicrystalline lattice is fabricated by additive manufacturing of Al-Cu-Fe icosahedral phase material.

21. A phi-graded structural element according to claim 1, wherein the element is integrated into a vehicle and the phi-graded zones are distributed along the vehicle length, such that the vehicle structure itself constitutes the multilayer parametric cascade medium.

22. The structural element of claim 21, wherein the vehicle length is divided into 13 sections with lengths in golden-ratio progression, and structural properties of each section are tuned to resonate at the phi-harmonic frequency corresponding to that section dimension.

23. An energy harvesting system comprising the structural element of claim 1, wherein the element is positioned in an environment having mechanical vibration exceeding 100 dB SPL and a thermal gradient exceeding 100 K, and the system extracts electrical energy from the thermoelectric electrode layers and directed acoustic energy from the element surfaces.

24. A method of generating a frequency comb in a solid-state medium, comprising:

(a) providing a quasicrystalline medium having zones with dimensions in golden-ratio progression;

(b) injecting broadband acoustic energy into the medium;

(c) allowing nonlinear difference-frequency mixing at zone boundaries to cascade energy from input frequencies to progressively lower frequencies along the phi-spaced resonant mode structure of the medium;

wherein the frequency comb output spans from the input frequency to an output frequency of f_{in} / phi^N where N is the total number of cascade steps supported by the medium geometry and nonlinear properties.

25. The method of claim 24, wherein the cascade spans at least 6 decades of frequency.

APPENDIX A: KEY PARAMETERS AND MATERIAL PROPERTIES

Parameter	Symbol	Value	Units
Golden ratio	phi	1.6180339887...	—
Golden angle	theta_g	137.508	degrees
QC layer count	N	13	—
Electrode positions	—	2, 8, 12	zone number
QC composition	—	Al63Cu25Fe12	—
QC hardness	HV	1000-1200	Vickers
QC friction coeff.	mu	0.04-0.06	—
CuNW Seebeck coeff.	S	80-90	uV/K
Steel sound velocity	c_steel	5,790	m/s
Air sound velocity	c_air	343	m/s
Steel acoustic impedance	Z_steel	46,320,000	Rayl
Air acoustic impedance	Z_air	420	Rayl
Engine acoustic power (33)	P_ac	3,574	MW
Exhaust temperature	T_hot	3,500	K
Cryo propellant temp	T_cold	90	K
Thermal gradient	delta-T	3,410	K
Carnot efficiency	eta_C	97.4	%
TE efficiency (35% Carnot)	eta_TE	34.1	%
Vehicle weight	W	49.05	MN
Grav. char. frequency	f_grav	0.3423	mHz
Cascade steps to f_grav	N_steps	~55-70	—